Q1: 16 March (Shift 1) - Numerical

Consider an arithmetic series and a geometric series having four initial terms from the set $\{11, 8, 21, 16, 26, 32, 4\}$. If the last terms of these series are the maximum possible four digit numbers, then the number of common terms in these two series is equal to______

Q2: 16 March (Shift 2) - Numerical

Let $\frac{1}{16}$, a and b be in G.P. and $\frac{1}{a}, \frac{1}{b}, 6$ be in A.P., where $a, b > 0$. Then $72(a+b)$ is equal to ____

Q3: 16 March (Shift 2) - Numerical

Let

$$\begin{aligned} S_n(x) = &\log_{a^{1/2}} x + \log_{a^{1/3}} x + \log_{a^{1/6}} x \\ &+ \log_{a^{1/1}} x + \log_{a^{1/18}} x + \log_{a^{1/27}} x + \ldots \end{aligned}$$

up to n -terms, where $a > 1$. If $S_{24}(x) = 1093$ and $S_{12}(2x) = 265$, then value of a is equal to ____

Q4: 17 March (Shift 2) - Numerical

If $1, \log_{10}(4^x - 2)$ and $\log_{10}\left(4^x + \frac{18}{5}\right)$ are in arithmetic progression for a real number x, then the value of the determinant $\begin{vmatrix} 2\left(x-\frac{1}{2}\right) & x-1 & x^2 \\ 1 & 0 & x \\ x & 1 & 0 \end{vmatrix}$ is equal to :

Q5: 18 March (Shift 1) - Single Correct

If α, β are natural numbers such that $100^\alpha - 199\beta = (100)(100) + (99)(101) + (98)(102) + \ldots\ldots + (1)(199)$, then the slope of the line passing through (α, β) and origin is :

(1) 540

(2) 550

(3) 530

(4) 510

Q6: 18 March (Shift 1) - Single Correct

$\frac{1}{3^2-1}+\frac{1}{5^2-1}+\frac{1}{7^2-1}+\ldots+\frac{1}{(201)^2-1}$ is equal to

(1) $\frac{101}{404}$

(2) $\frac{25}{101}$

(3) $\frac{101}{408}$

(4) $\frac{99}{400}$

Q7: 18 March (Shift 2) - Single Correct

Let S_1 be the sum of first 2 n terms of an arithmetic progression. Let S_2 be the sum of first 4n terms of the same arithmetic progression. If $(\mathrm{S}_2-\mathrm{S}_1)$ is 1000, then the sum of the first 6n terms of the arithmetic progression is equal to:

(1) 1000

(2) 7000

(3) 5000

(4) 3000

Q8: 18 March (Shift 2) - Numerical

If $\sum_{r=1}^{10} r!\left(r^3+6r^2+2r+5\right)=\alpha(11!)$, then the value of α is equal to ___

Answer Key

Q1 (3)	**Q2** (14)	**Q3** (16)	**Q4** (2)
Q5 (2)	**Q6** (2)	**Q7** (4)	**Q8** (160)

Q1 (3)

GP : 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048,

4096,8192

AP : 11,16,21,26,31,36

Common terms : 16,256,4096 only

Q2 (14)

$a^2 = \frac{b}{16} \Rightarrow \frac{1}{b} = \frac{1}{16a^2}$

$\frac{2}{b} = \frac{1}{a} + 6$

$\frac{1}{8a^2} = \frac{1}{a} + 6$

$\frac{1}{a^2} - \frac{8}{a} - 48 = 0$

$\frac{1}{a} = 12, -4 \Rightarrow a = \frac{1}{12}, -\frac{1}{4}$

$a = \frac{1}{12}, a > 0$

$b = 16a^2 = \frac{1}{9}$

$\Rightarrow \quad 72(a+b) = 6 + 8 = 14$

Q3 (16)

$\mathrm{S_n(x)} = (2 + 3 + 6 + 11 + 18 + 27 + \ldots\ldots + \mathrm{n - terms}) \log_\mathrm{a} \mathrm{x}$

Let $\mathrm{S_1} = 2 + 3 + 6 + 11 + 18 + 27 + \ldots + \mathrm{T_n}$

$\mathrm{S_1} = 2 + 3 + 6 + \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots + \mathrm{T_n}$

$\mathrm{T_n} = 2 + 1 + 3 + 5 + \ldots\ldots + \mathrm{n\ terms}$

$\mathrm{T_n} = 2 + (\mathrm{n} - 1)^2$

$\mathrm{S_1} = \Sigma \mathrm{T_n} = 2\mathrm{n} + \frac{(\mathrm{n}-1)\mathrm{n}(2\mathrm{n}-1)}{6}$

$\Rightarrow \mathrm{S_n(x)} = \left(2\mathrm{n} + \frac{\mathrm{n}(\mathrm{n}-1)(2\mathrm{n}-1)}{6}\right) \log_\mathrm{a} \mathrm{x}$

$\mathrm{S_{24}(x)} = 1093 \quad (\text{ Given })$

$\log_\mathrm{a} \mathrm{x} \left(48 + \frac{23.24.47}{6}\right) = 1093$

$\log_\mathrm{a} \mathrm{x} = \frac{1}{4} \quad \ldots.(1)$

$\mathrm{S_{12}(2x)} = 265$

$S_{12}(2x) = 265$

$\log_a 2x = \frac{1}{2} \ldots (2)$

$(2) - (1)$

$\log_a 2x - \log_a x = \frac{1}{4}$

$\log_a 2 = \frac{1}{4} \Rightarrow a = 16$

Q4 (2)

$2\log_{10}(4^x - 2) = 1 + \log_{10}\left(4^x + \frac{18}{5}\right)$

$(4^x - 2)^2 = 10\left(4^x + \frac{18}{5}\right)$

$(4^x)^2 + 4 - 4(4^x) - 32 = 0$

$(4^x - 16)(4^x + 2) = 0$

$4^x = 16$

$x = 2$

$\begin{vmatrix} 3 & 1 & 4 \\ 1 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix} = 3(-2) - 1(0-4) + 4(1)$

$= -6 + 4 + 4 = 2$

Q5 (2)

$S = (100)(100) + (99)(101) + (98)(102) \ldots\ldots. (2)(198) + (1)(199)$

$S = \sum_{x=0}^{99}(100 - x)(100 + x) = \sum 100^2 - x^2$

$= 100^3 - \frac{99 \times 100 \times 199}{6}$

$\alpha = 3 \quad \beta = 1650$

$\text{slope} = \frac{1650}{3} = 550$

Q6 (2)

$T_n = \frac{1}{(2n+1)^2 - 1} \frac{1}{(2n+2)2n} = \frac{1}{4(n)(n+1)}$

$= \frac{(n+1)-n}{4n(n+1)} = \frac{1}{4}\left(\frac{1}{n} - \frac{1}{n+1}\right)$

$S = \frac{1}{4}\left(1 - \frac{1}{101}\right) = \frac{1}{4}\left(\frac{100}{101}\right) = \frac{25}{101}$

Q7 (4)

$S_{2n}=\frac{2n}{2}[2a+(2n-1)d], S_{4n}=\frac{4n}{2}[2a+(4n-$

$1)\mathrm{d}]$

$\Rightarrow \mathrm{S}_2-\mathrm{S}_1=\frac{4\mathrm{n}}{2}[2\mathrm{a}+(4\mathrm{n}-1)\mathrm{d}]-\frac{2\mathrm{n}}{2}[2\mathrm{a}+(2\mathrm{n}-$

$1)\mathrm{d}]=4\mathrm{an}+(4\mathrm{n}-1)2\mathrm{nd}-2\mathrm{na}-(2\mathrm{n}-1)\mathrm{dn}$

$=2na+nd[8n-2-2n+1]$

$\Rightarrow 2\mathrm{na}+2\mathrm{n}[6\mathrm{n}-1]=1000$

$2a+(6n-1)d=\frac{1000}{n}$

Now, $S_{6n}=\frac{6n}{2}[2a+(6n-1)d]$

$=3\mathrm{n}\cdot\frac{1000}{\mathrm{n}}=3000$

Q8 (160)

$\sum_{r=1}^{10}\mathrm{r}!\{(\mathrm{r}+1)(\mathrm{r}+2)(\mathrm{r}+3)-9(\mathrm{r}+1)+8\}$

$=\sum_{\mathrm{r}=1}^{10}[\{(\mathrm{r}+3)!-(\mathrm{r}+1)!\}-8\{(\mathrm{r}+1)!-\mathrm{r}!\}]$

$=(13!+12!-2!-3!)-8(11!-1)$

$=(12.13+12-8)\cdot 11!-8+8$

$=(160)(11)!$

Hence $\alpha=160$